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WANDERLINK TRAVEL — SYSTEM ARCHITECTURE DOCUMENT

Document Version: 1.0

Project Name: WanderLink Travel

Target Domain: wanderlinktravel.com

Primary Stack: Next.js (App Router), React 19, TypeScript, Tailwind CSS, Vercel, OpenAI API, Unsplash API

1. Executive Summary & System Scope

WanderLink Travel is an AI-native travel recommendation platform designed to eliminate destination discovery friction. Users complete a 2-minute "Travel DNA" questionnaire covering budget thresholds, group composition, duration, and climate preferences.

The platform uses OpenAI models to compute recommendation vectors and match score ratings, enrich destination cards with contextual high-resolution imagery via the Unsplash API, and output direct booking paths to partner monetization networks (Booking.com, Expedia Group, Viator, GetYourGuide, Airalo).

Core Architectural Objectives

- **Sub-2-Second Latency:** Ultra-fast API response times for questionnaire submission to recommendation rendering.
 - **Zero-Infrastructure Overhead:** Fully serverless deployment pipeline leveraging Vercel, Next.js App Router, and Edge Networks.
 - **High Conversion UX:** Fully responsive, accessible, dark/light theme supporting mobile-first travel search.
 - **FinOps Governance:** Strict rate-limiting, edge caching, and dynamic token allocation to keep API operational costs under \$0.005 per user interaction.
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2. Infrastructure Architecture

WanderLink’s infrastructure leverages a globally distributed, serverless cloud model managed primarily via Vercel, GitHub, and Cloudflare.

System Infrastructure Matrix

Layer	Component / Technology	Specification & Configuration
DNS & Edge Proxy	Cloudflare	Anycast DNS, SSL/TLS termination, DDoS Mitigation, Edge Security.
Hosting & Compute	Vercel Serverless Functions	Node.js Serverless runtime (AWS Lambda backbone under Vercel hood), Auto-scaling.
Deployment Pipeline	Vercel Git Integration	Push-to-deploy triggered by git push origin main on GitHub repository (BanerjeeArindam/wanderlink).
Asset Storage	Public Asset CDN (/public)	Static distribution of vector logos (logo.svg), avatars (profile-logo.png), and metadata assets.

3. Application Architecture

The application is built on the **Next.js App Router** framework:

```
app/  
└── layout.tsx    # Global Shell: Navbar, Footer, SVG Logos, Impact Tag
```

```
|— page.tsx      # Home Hero, Ambient Animated Visuals, Feature Grid
|— questionnaire/
|   |— page.tsx  # 8-Step Interactive Form UI
|— results/
|   |— page.tsx  # Recommendation Grid & Booking Route Linker
|— api/
|   |— recommend/
|       |— route.ts # Serverless Endpoint: OpenAI Engine + Unsplash Image Fetching
```

Application Flow & Data Sequence

1. **Client-Side Questionnaire (app/questionnaire/page.tsx):** Captures user signals (group size, budget, preferred climate, activities, duration).
2. **API Endpoint Execution (app/api/recommend/route.ts):** Configured with export const dynamic = 'force-dynamic' to ensure fresh execution without static pre-rendering build errors. Instantiates OpenAI client lazily within the HTTP POST context to protect API credentials.
3. **External API Calls:** Prompt submitted to gpt-4o-mini, cleaned JSON parsed, parallel image query dispatched to Unsplash API using Promise.all().
4. **UI Render (app/results/page.tsx):** Renders enriched cards containing destination titles, match scores, taglines, reasons to visit, key highlights, and pre-formatted affiliate link parameters.

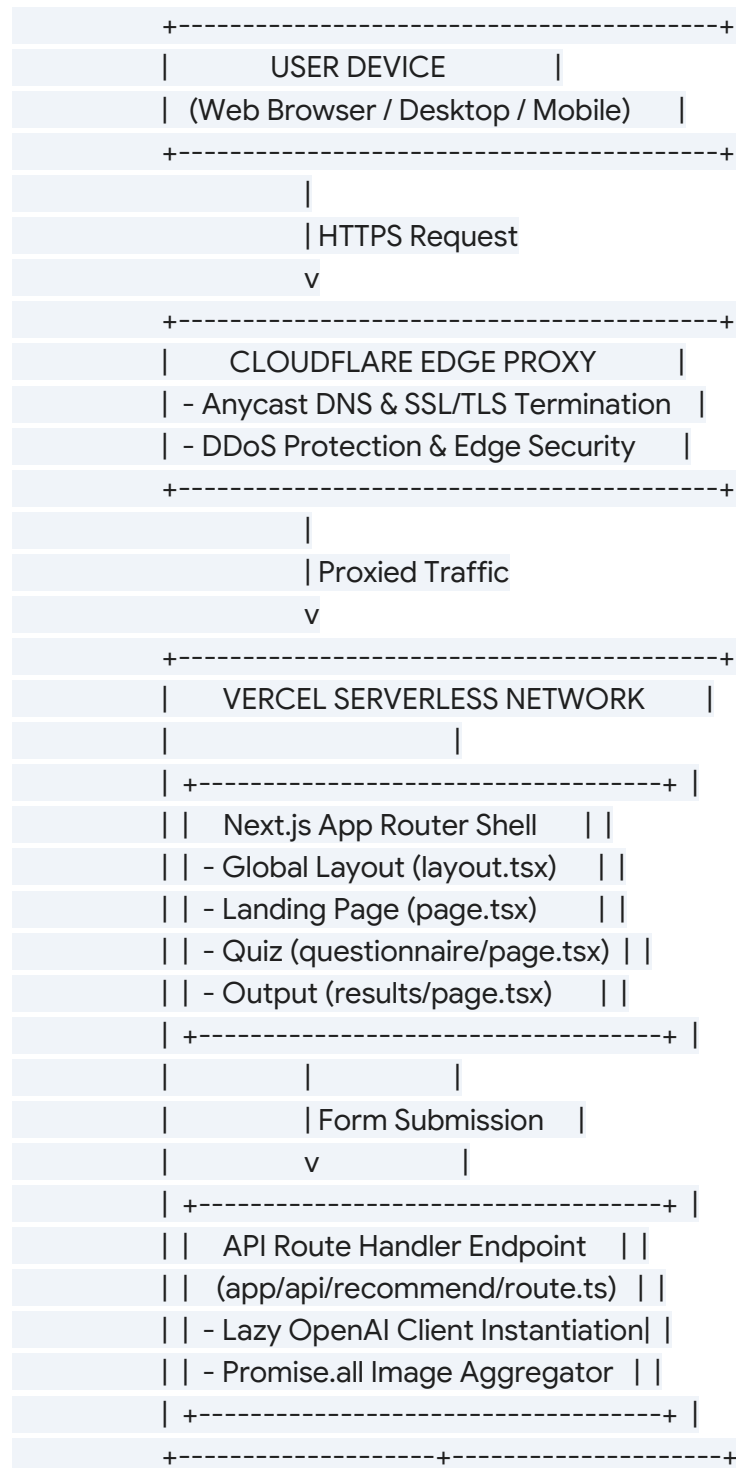
4. Capability Map

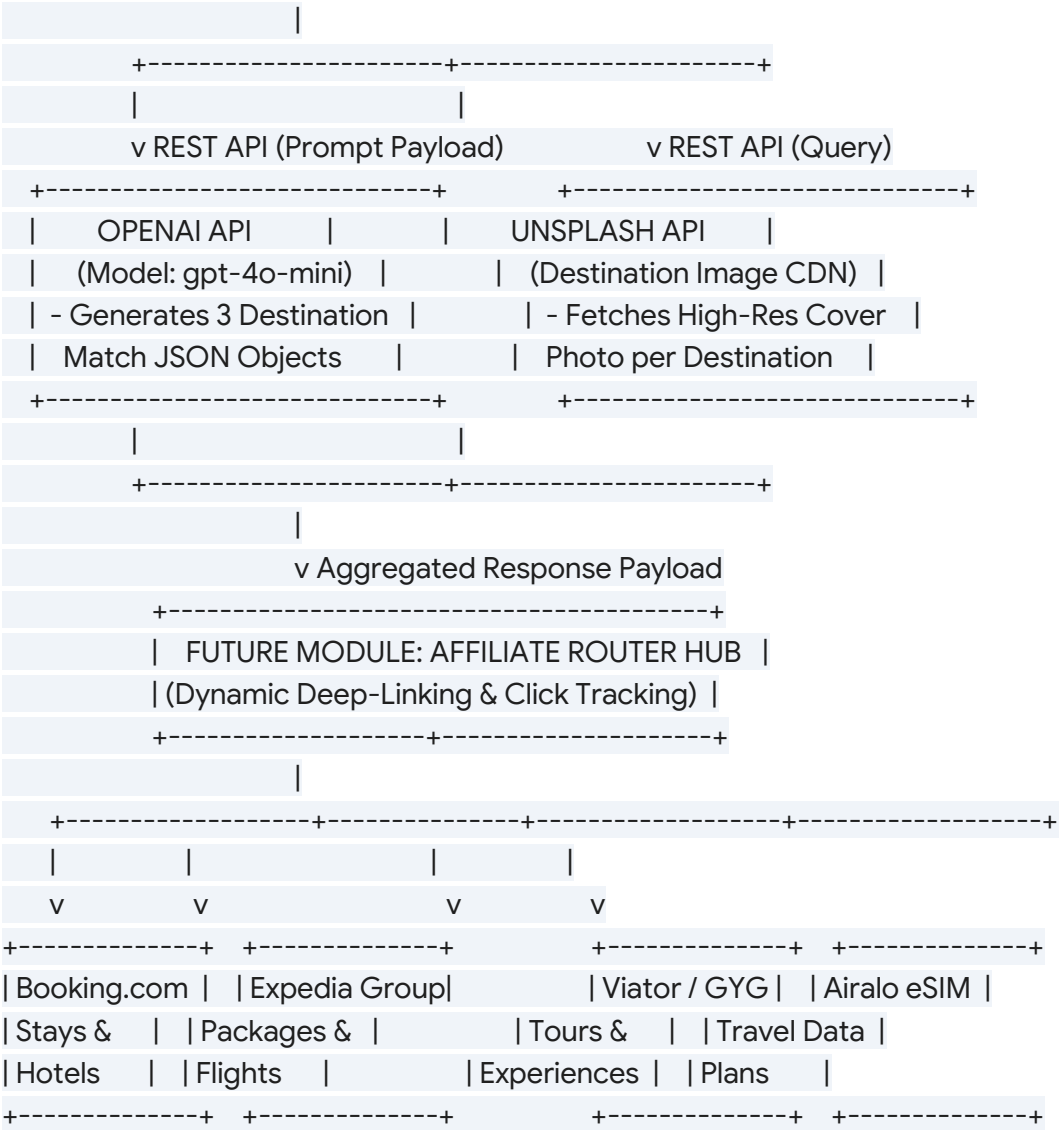
WANDERLINK CAPABILITY MAP		
1. CUSTOMER ENGAGEMENT & DISCOVERY		
1.1 Travel DNA Assessment UI	— Responsive 8-step questionnaire for parameter capturing	
1.2 Brand Landing Experience	— High-conversion hero, ambient themes, & visual showcase	
1.3 Social Media Marketing Assets	— Multi-channel assets for Facebook, Instagram, & Pinterest	

2. INTELLIGENCE & MATCHING ENGINE	
2.1 Prompt Engineering & Structuring	OpenAI System Prompts enforcing strict JSON output schemas
2.2 Recommendation Scoring	Multi-variable alignment (Budget, Climate, Group Vibe)
2.3 Visual Context Enrichment	Real-time image fetching via Unsplash REST API
3. AFFILIATE BOOKING ROUTER & MONETIZATION HUB (FUTURE MODULE)	
3.1 Accommodations Routing Module	Dynamic URL generation & deep-linking for Booking.com & Expedia with tracking IDs (`aid` / `camref`)
3.2 Tour & Activity Linking Module	Tour search query deep-links for Viator & GetYourGuide
3.3 Connectivity & eSIM Module	eSIM referral routing via Airalo / Impact.com
3.4 Outbound Click & Revenue Tracking	Serverless event logging to track conversion metrics and outbound affiliate click-through rate (CTR)
4. DEVOPS & CI/CD INFRASTRUCTURE	
4.1 Continuous Deployment	Git-driven production pipelines (GitHub → Vercel)
4.2 Edge Execution & Routing	Anycast DNS & Edge function runtime via Cloudflare & Vercel
4.3 AI-Native Developer Tooling	Cursor IDE & Builder.io visual layout synchronization
5. FINOPS & GOVERNANCE	
5.1 Token Budget Protection	Hard spend limits (\$10–\$20 cap) on OpenAI API Usage
5.2 Serverless Cost Management	On-demand execution with zero idle compute overhead
5.3 API Rate Limit Guardrails	Fallback handlers for API quota breaches and timeouts

5. Visual Architecture Specifications

Component Architecture & Data Flow Diagram





6. Data Architecture & Schema Definition

WanderLink follows a **stateless, privacy-first data architecture**. No personally identifiable information (PII) is stored in persistent databases.

User Preference Input Payload (Client to Server)

TypeScript

```
interface TravelPreferences {
  groupType: 'solo' | 'couple' | 'family' | 'friends';
  budgetLevel: 'budget' | 'moderate' | 'luxury';
  durationDays: number;
  travelMonth: string;
  preferredClimate: 'tropical' | 'mild' | 'cold' | 'any';
  activityVibe: string[];
  visaPreference?: 'visa-free' | 'any';
}
```

Enriched Destination Output Schema (Server to Client)

TypeScript

```
interface RecommendationCard {
  destination: string; // e.g., "Kyoto, Japan"
  countryCode: string; // e.g., "JP"
  matchScore: number; // e.g., 96
  heroTagline: string; // Short narrative headline
  reasonsToVisit: string[]; // 3 custom bullet points
  keyHighlights: string[]; // 3 key attractions
  affiliateQuery: string; // Sanitized search term for affiliate routing
  imageUrl: string; // Direct Unsplash CDN photo URL
}
```

7. Operational Architecture & Maintenance

Development Environment & Tooling

- **Primary IDE:** Cursor AI / VS Code with integrated Git source control.
- **Local Development Server:** Node.js (v24.x LTS) with npm run dev running at http://localhost:3000.
- **Version Control Pipeline:**

```
Bash
git add .
git commit -m "feat: description of update"
git push origin main # Triggers automatic Vercel production build
```

Monitoring & Incident Handling

- **Vercel Runtime Logs:** Real-time console error tracing for app/api/recommend/route.ts.
- **OpenAI Usage Dashboard:** Monitoring token usage, error rates, and API key billing limits.
- **Failover Fallbacks:** Unsplash requests fall back to a high-res default cover photo (https://images.unsplash.com/photo-1488646953014-85cb44e25828) if rate limits are reached.

8. FinOps & Cost Optimization Strategy

Cost Breakdown Model (Per 1,000 Recommendations)

- **Vercel Hosting:** \$0.00 (Covered under Free Tier)
- **Unsplash API:** \$0.00 (Free Tier: 50 requests/hr)
- **OpenAI GPT-4o-mini:** ~\$0.15 - \$0.30 (500 input / 300 output tokens per request)
- **Estimated Revenue Potential:** \$20.00 - \$100.00+ (Via affiliate conversions)

FinOps Safeguards

1. **Model Selection:** gpt-4o-mini cuts token costs by ~95% compared to gpt-4o.
2. **Hard Billing Caps:** Monthly spending capped at \$10-\$20 USD via the [OpenAI Billing Dashboard](#).
3. **Serverless Idle Protection:** Zero compute charges during idle traffic periods.